

Nagar Nigam Kiosk System

End-to-End System Workflow & Execution Architecture Guide

This document provides a comprehensive, structured workflow explanation of the **Nagar Nigam Citizen Service Kiosk System**. It maps out the end-to-end user journey, administrative system oversight, and back-end database execution workflows for internal review and management analysis.

System Architecture Note: The platform is split into two operating modules: the customer-facing **Citizen Service Kiosk Interface** and the secure back-end **Administrative Management Portal**.

Module 1: Citizen Service Kiosk Interface

This module guides a citizen through pulling official documents from government databases, processing transactions via digital payment frameworks, or requesting data modifications.

Workflow A: Certificate Printing Service

- Language Selection:** The citizen lands on a bilingual touch home screen and selects their preferred operational language (**Hindi** or **English**) to initiate the user flow.
- Service Classification:** The portal offers two discrete system routing operations:
 - Print Certificate (प्रमाणपत्र प्रिंट करवाएं)
 - Correct Certificate (प्रमाणपत्र सुधारें)
- Portal Redirection:** Upon selecting "Print Certificate," the terminal automatically connects with external verification networks. Clicking "**Search in Pehchan Portal**" triggers a secure automated handoff script to the official *Rajasthan Civil Registration System (Pehchan Portal)*.
- Document Retrieval:** The kiosk pulls the requested digital document into a secure sandboxed web session. The document is cached dynamically inside the local system framework at `nagar-nigam-org/server/temp/downloads`.
- Form Logging & Copy Selection:** The user is returned to the core kiosk UI to record transaction criteria:
 - Applicant Name
 - 10-Digit Mobile Number
 - Registration Number
 - Certificate Type (e.g., Birth Certificate)
 - Number of Copies (Selectable range from 1 to 5; unit cost is calculated at a base rate of ₹20 per printed sheet).

6. **Integrated Payment Gateway:** The system builds a dynamic itemized invoice outlining total sheet volume and billing calculations. A **dynamic UPI QR Code** bound directly to the terminal's target address (`nagarnigam.kiosk@sbi`) is generated on screen.
7. **Hardware Print Execution:** When payment receipt is verified by the server, the system fires a background loading spooler. This invokes the native hardware print execution wrapper, mapping the document into target alignment matrices to produce the physical print out.

Workflow B: Certificate Correction Request (Queue Token Generation)

1. **Initiation:** From the main interface dashboard, the citizen registers a modification intent by choosing **"Correct Certificate."**
2. **Field Categorization:** The user specifies which database field requires modification (e.g., *Gender, Date of Birth, Name string corrections, or Parentage fields*).
3. **Data Modification Input:** The application loads a secure data mutation layout grid. The user inputs their baseline registration numbers and updates parameters across explicit **"Old Value"** and **"New Value"** text blocks.
4. **Processing Token Payment:** A flat nominal processing token fee of ₹50 is calculated. The user scans an automated UPI QR code layout to authorize the modification transaction.
5. **Token Handshake & Print:** Upon successful authorization, the local kiosk outputs a physical sequentially incremented service token (e.g., **TKN-114, Counter 1**) and completely flushes memory caches to reset back to a clean landing layout.

Module 2: Administrative Management Portal

Accessible by local administrators via specialized system loops, this interface serves as the primary data control room for terminal metrics, system operational health, and tracking digital audits.

1. Live Analytical Dashboard

- **Real-Time Core Metrics:** Displays cumulative counters capturing absolute volume parameters across *Printed Certificates, Active Correction Tokens*, and total *Collected Kiosk Revenue*.
- **System Diagnostics Panel:** A hardware checker continually samples internal connection heartbeats to flag system status parameters instantly as **ONLINE**.
- **Live Diagnostic Logs Stream:** A terminal stream output maps live structural events, recording active server bindings on port `5000`, database handshakes, session parameters, and transaction records.

2. Queue Manager & Text-To-Speech Framework

- **Active Status Matrix:** A tracking screen renders the current token number active at the teller desks.

- **Automated Audio Voice Engine:** Clicking "**Announce Current**" runs a built-in Text-to-Speech (TTS) audio layer that broadcasts queue calls clearly across the waiting lobby (e.g., "*Token number TKN-101, please proceed to Counter 1*").
- **Queue Progression Controls:** Local office team members manipulate client routing via dedicated "**Call Next Token**" triggers or globally flush token queues via a secure "**Reset System**" option.

3. Relational Database Viewer & Audit Ledgers

An administrative table filtering engine lets management access raw database schemas directly to inspect records across specific tracking buckets:

- **Admin / SuperAdmin Table:** Registers identity rows mapping authorized personnel names, official organization emails, and encrypted validation passwords.
- **Payment Ledger:** Tracks all electronic financial interactions, binding unique payment hashes to registration metadata, transaction weights, status strings, and time signatures.
- **Certificate Print Record:** Logs historical verification tracking details, file names, volume numbers, and applicant indices.
- **Counter Correction Record:** A highly granular diagnostic table tracking database modifications. It captures nested array datasets that map precise changes made (e.g., recording the modification of an incorrect value from 'male' to 'female').
- **Token Lifecycle Schema:** Records entry time tracking data, assigned kiosk stations, and current operation states (*WAITING / SERVING / COMPLETED*).